UNIT 1 MINI-PROJECT REPORT

Enterprise Student Record and Academic Management Platform

Design Thinking and Product Planning

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Unit 1 Mini-Project Report

Enterprise Student Record and Academic Management Platform

Subject area: Design Thinking and Product Planning

1. Introduction

Educational institutions manage large volumes of student and academic information. A structured digital platform can help keep records consistent, improve access to information, and reduce repetitive administrative work. This mini-project applies product engineering and design-thinking methods before beginning full application development.

2. Business Understanding

Current process

Student information may be maintained across admission forms, spreadsheets, attendance registers, examination records, and separate report files. Staff may need to enter the same details repeatedly and reconcile inconsistencies manually.

Main problems

- Duplicate or incomplete student records
- Slow search and retrieval of student information
- Manual attendance summaries
- Delays in preparing examination and academic reports
- Limited visibility into missing or inconsistent records
- Unclear access control when files are shared informally

Proposed business outcome

Provide a single, controlled source of academic records with role-based access and consistent workflows for registration, attendance, results, and reporting.

3. Product Vision

Vision statement: To provide a secure, reliable, and easy-to-use academic management platform that helps higher education institutions maintain accurate student records and deliver timely academic information.

Mission: Simplify academic administration by organizing student profiles, course registration, attendance, examination results, and reports in one platform.

Business objectives

1. Reduce repeated data entry and duplicate records.
2. Make student information easier for authorized users to find.
3. Improve consistency of attendance and examination records.
4. Produce routine academic reports more efficiently.
5. Protect student information through role-based access and audit trails.

4. Stakeholder Analysis

5. User Personas

Persona A: Aditi — Student
Goals: Check registered courses, attendance percentage, and published results.

Challenges: Information is spread across notices, spreadsheets, and office visits.

Needs: A simple dashboard, accurate records, and timely status updates.

Success indicator: Aditi can find her attendance and published results without asking staff to search manual files.

Persona B: Mr. Patil — Faculty Member
Goals: View assigned classes, record attendance, and submit marks.

Challenges: Repeated manual entry and difficulty correcting mistakes consistently.

Needs: Class-wise lists, input validation, save confirmation, and an authorized correction workflow.

Success indicator: Attendance and marks can be entered and reviewed with fewer avoidable errors.

Persona C: Ms. Deshmukh — Academic Administrator
Goals: Maintain student records and prepare department reports.

Challenges: Duplicate records, incomplete fields, and time-consuming compilation.

Needs: Search, validation, role permissions, and report filters.

Success indicator: Routine student and academic reports can be generated from consistent records.

6. Design Thinking

Empathize
Understand the daily tasks of students, faculty, administrators, examination staff, and IT support. Gather feedback through interviews, observation, and short surveys before finalizing detailed workflows.

Define
Problem statement: Students and institutional staff need a dependable way to manage and access academic information because fragmented manual records make updates, attendance tracking, result preparation, and reporting slow and error-prone.

Ideate
Potential solutions:

- Centralized student profile
- Course registration and class lists
- Attendance entry with validation
- Examination marks and controlled result publication
- Search and filtered reports
- Role-based permissions and audit history
- Backup and recovery procedures

Prototype
Create low-fidelity wireframes for:

6. Login and role-specific dashboard
7. Student profile and course registration
8. Faculty attendance screen
9. Examination marks and result screen
10. Administrator report dashboard

Test
Ask representative users to complete tasks using the prototype. Observe whether they can find a student, register a course, record attendance, and locate a published result. Record confusion points and revise the design. No user testing is claimed as completed in this planning document; it is a planned activity.

7. IBM Design Thinking

Focus on user outcomes
Measure whether students can find published academic information easily and whether staff can complete routine record tasks with fewer errors.

Restless reinvention
Use feedback, issue logs, and task measurements after each iteration to improve workflows rather than treating the first design as final.

Diverse empowered teams
Include students, faculty, academic administration, examination staff, developers, testers, and IT operations in discovery and review.

8. IBM Loop: Observe → Reflect → Make
- Observe: Review user feedback, task completion issues, missing data patterns, and support requests.
- Reflect: Identify root causes, compare them with project goals, and prioritize improvements.
- Make: Update the prototype or product backlog, implement the highest-value improvement, and test it.

Repeat the cycle throughout development.

9. Business Requirements Document (BRD)

Business need
Create a centralized system for accurate, accessible, and controlled student academic records.

Scope included
- Student profile and registration records
- Course registration
- Attendance records
- Examination marks and published results
- Academic reports
- Role-based access and audit history

Out of scope for the first release
- Fee payment gateway
- AI-based predictions

- Biometric attendance devices
- Integration with external government or university systems unless approved later
- Native mobile applications

Assumptions

- The institution will define approved academic rules and user roles.
- Authorized staff will maintain accurate source information.
- Users will have access to a supported browser and institutional network or internet.
- The institution will approve retention, backup, and privacy policies.

Constraints and risks

- Personal academic data must be protected.
- Incorrect source data may create incorrect reports.
- User adoption requires training and feedback.
- Integration and hosting costs require approval.
- Result publication must be controlled and auditable.

Acceptance criteria

- Authorized users can locate a student record.
- Course, attendance, and result records are associated with the correct student and academic period.
- Unauthorized roles cannot edit restricted records.
- Reports show the selected filters and data period.
- Changes to sensitive academic records are auditable.

10. Functional Requirements Specification (FRS)

11. Non-Functional Requirements Specification (NFRS)

12. Minimum Viable Product (MVP)

The first release should include:

11. Login and role-based access
12. Student profile management
13. Course and semester records
14. Course registration
15. Attendance entry and viewing
16. Examination marks and controlled result publication
17. Basic student, attendance, and result reports

Deferred: payment integrations, advanced analytics, AI features, biometric devices, and external-system integrations.

13. Product Roadmap

14. Initial System Architecture

Conceptual layers

18. User interface: student, faculty, administrator, and examination dashboards.
19. Application services: authentication, student records, course registration, attendance, examination, and reporting.
20. Data layer: relational data model with integrity constraints and backups.
21. Operations layer: version control, automated tests, CI/CD, deployment configuration, logs, and monitoring.

See `../diagrams/system-architecture.svg`.

15. Project Charter

Project title: Enterprise Student Record and Academic Management Platform

Purpose: Plan a centralized academic record solution through business analysis and design thinking.

Sponsor: To be nominated by the institution.

Project lead: Student project team (names to be added by the student).

Users: Students, faculty, academic administrators, examination staff, IT support.

Deliverables: BRD, vision, stakeholder analysis, personas, FRS, NFRS, design-thinking report, IBM Design Thinking report, IBM Loop documentation, MVP, roadmap, architecture concept, repository structure.

Success criteria: All Unit 1 deliverables are documented, reviewed against the assignment checklist, and organized in a Git repository.

Dependencies: Access to stakeholder feedback, approved academic rules, and faculty review.

Risks: Incomplete requirements, privacy concerns, changing academic rules, and limited time for prototype validation.

Approval: Faculty guide / instructor (name and date to be filled in).

16. Milestone Plan

17. Product Discovery Plan

Questions for discovery:

- Which student record tasks take the most time?
- Which fields are mandatory for student registration?
- Who is permitted to edit attendance and examination records?
- How are corrections approved and recorded?
- When and how are results published?
- Which reports are required by each department?
- What privacy, retention, backup, and access policies apply?

Methods: short interviews, workflow observation, process mapping, prototype walkthroughs, and review of existing forms with permission.

18. Conclusion

The proposed Student Record and Academic Management Platform addresses fragmented academic information through a centralized, role-based product concept. The project follows the assignment's focus on

business understanding, design thinking, stakeholder analysis, requirements, product vision, MVP planning, roadmap, and repository initialization. The next step is to validate the assumptions with real stakeholders and develop the product incrementally in later units.

Requirements Summary Tables

Functional Requirements

ID	Requirement	Priority
FR-01	Authenticate users and enforce role-based permissions	Must
FR-02	Authorized administrators manage student records	Must
FR-03	Validate required fields and prevent duplicate student IDs	Must
FR-04	Maintain course and semester information	Must
FR-05	Register students for approved courses	Must
FR-06	Record and review attendance for assigned classes	Must
FR-07	Enter and review examination marks	Must
FR-08	Publish results under authorized control	Must
FR-09	Generate student, attendance, and result reports	Must
FR-10	Record audit events for sensitive changes	Should
FR-11	Search and filter records	Must
FR-12	Document backup and restore procedure	Should

Non-Functional Requirements

ID	Requirement
NFR-01	Security: least-privilege access, secure authentication, protected secrets, encrypted transport.
NFR-02	Privacy: show only information required for a user's role.
NFR-03	Usability: consistent labels, clear validation, responsive layouts.
NFR-04	Performance: target common searches under 2 seconds under agreed test load; verify by testing.
NFR-05	Reliability: validate relationships between students, courses, attendance, and results.
NFR-06	Availability: define target with institution and monitor it.
NFR-07	Maintainability: separate interface, business logic, and data access.
NFR-08	Scalability: verify capacity through load testing.
NFR-09	Auditability: record who changed sensitive data and when; never log secrets.
NFR-10	Recovery: document and test backup restoration.

Project Folder Structure

```
Student_Record_Academic_Management_Platform/
├── .github/workflows/      # Planned CI workflow
```

```
├── app/
│   ├── modules/          # Future application modules
│   ├── database/         # Future schema and migration files
│   └── tests/            # Future automated tests
├── diagrams/             # Architecture diagram
├── docs/
│   ├── PROJECT_REPORT.md
│   └── PROJECT_FOLDER_STRUCTURE.md
├── docker/               # Future container configuration
├── kubernetes/           # Future deployment manifests
├── terraform/            # Future infrastructure as code
├── ansible/              # Future configuration playbooks
├── monitoring/           # Future monitoring configuration
├── scripts/              # Helper scripts
├── .gitignore
└── README.md
```

Folders for application code and deployment tools are initialized for future units; this Unit 1 submission is primarily planning documentation and does not claim these components are implemented.

System Architecture Diagram

The diagram is included separately in the repository at diagrams/system-architecture.svg.

Presentation layer → Application services → Data layer, supported by operations, testing, monitoring, security, and privacy.

Submission Checklist

☐ Business Requirements Document (BRD)

☐ Product Vision Document

☐ Stakeholder Analysis Report

☐ User Persona Report

☐ Functional Requirements Specification (FRS)

☐ Non-Functional Requirements Specification (NFRS)

☐ Design Thinking Report

☐ IBM Design Thinking Report

☐ IBM Loop Documentation

☐ MVP Definition

☐ Product Roadmap

☐ Initial System Architecture Diagram

☐ Project Charter

☐ Project Repository Initialization

☐ Project Folder Structure

☐ Product Discovery Documentation